Dr. A.P.J. Abdul Kalam Technical University, Lucknow

Course Code: KCS-401 • Course Title: OPERATING SYSTEMS

Type: Core Theory • Teaching Scheme: 3-1-0 • Credits: 4

Unit / Module	Topics & Subtopics	Hours
UNIT 1: Introduction	• Operating System Overview: Functions, Evolution, Batch, Interactive, Time-Sharing, Real-Time Systems • OS Architecture: Layered Structure, System Components, System Calls, Operating System Services • Kernel Design: Monolithic Kernels, Microkernel Systems, Reentrant Kernels	**[08 Hours]**
UNIT 2: Concurrent Processes	• Process Concepts: Principle of Concurrency, Process Generation, Process States • Synchronization: Producer-Consumer Problem, Critical Section Problem, Dekker's Solution, Peterson's Algorithm • Semaphores: Counting Semaphores, Binary Semaphores, Test-and-Set Operation • Classical Problems: Dining Philosophers Problem, Sleeping Barber Problem • Inter-Process Communication: Message Passing, Shared Memory	**[08 Hours]**
UNIT 3: CPU Scheduling & Deadlocks	• Scheduling Concepts: Performance Criteria, Schedulers, Process Control Block (PCB) • Scheduling Algorithms: First-Come First-Served (FCFS), Shortest Job First (SJF), Round Robin (RR), Priority Scheduling • Threads: Thread Models, Multithreading, Multiprocessor Scheduling • Deadlocks: System Model, Deadlock Characterization, Deadlock Prevention, Deadlock Avoidance (Banker's Algorithm), Deadlock Detection and Recovery	**[08 Hours]**
UNIT 4: Memory Management	• Memory Allocation: Fixed Partitioning, Variable Partitioning, Compaction, Protection Schemes • Virtual Memory: Paging, Page Table Structures, Segmentation, Paged Segmentation • Page Replacement: FIFO, Optimal, Least Recently Used (LRU), Thrashing and Working Set Model • Cache Memory: Cache Organization, Locality of Reference	**[08 Hours]**
UNIT 5: I/O & File Systems	• I/O Management: I/O Devices, Device Drivers, I/O Buffering, Direct Memory Access (DMA) • Disk Scheduling: FCFS, SSTF, SCAN, C-SCAN, LOOK, RAID Architectures • File Systems: File Concepts, Directory Structures, Allocation Methods (Contiguous, Linked, Indexed) • Protection: File Sharing, Access Matrix, Security Mechanisms	**[08 Hours]**